



MRC
Biostatistics
Unit



UNIVERSITY OF
CAMBRIDGE

Bayesian Methods for Clinical Trials

Lecture 1: Clinical trials and the role of statistics

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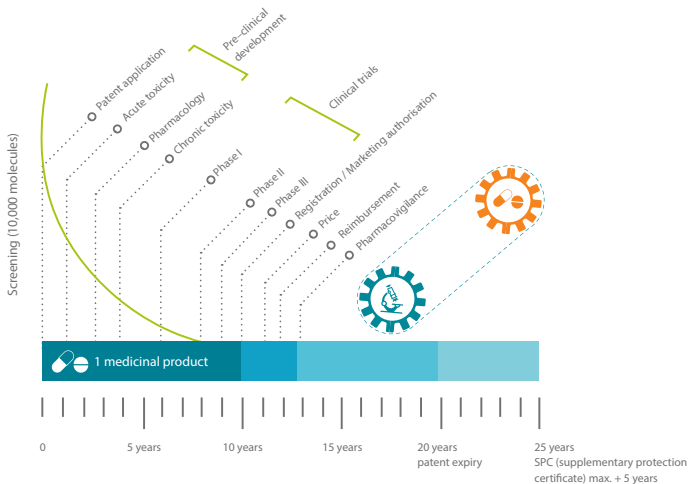
MRC Biostatistics Unit

Ben Goldacre, Guardian 1-09-08:

- **Before 1935** doctors were basically useless
- **1935-1995:** - antibiotics, dialysis, transplants, intensive-care units, heart surgery, every drug you've ever heard of
- **1995-now:** - the low-hanging fruit of medical research has all been harvested, and the industry is rapidly running out of new drugs

The development process

PHASES OF THE RESEARCH AND DEVELOPMENT PROCESS



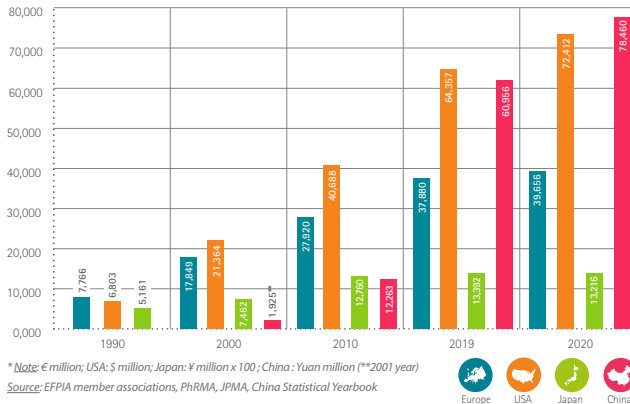
Source: European Federation of Pharmaceutical Industries and Associations (2022)

Development of a novel medicinal product

- takes 10-15 years
- costs several hundred million euros on average
 - ▶ largest contributors are confirmatory (phase III) trials
 - ▶ often involve thousands of patients with follow-up period frequently lasting years

Cost on R&D in Pharmaceutical industry

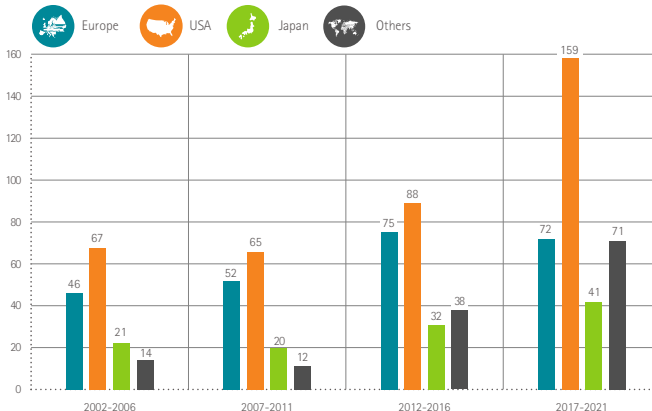
PHARMACEUTICAL R&D EXPENDITURE IN EUROPE, USA, JAPAN AND CHINA
(MILLION OF NATIONAL CURRENCY UNITS)*, 1990–2020



Source: European Federation of Pharmaceutical Industries and Associations (2022)

New molecular entities

NUMBER OF NEW CHEMICAL AND BIOLOGICAL ENTITIES (2002-2021)

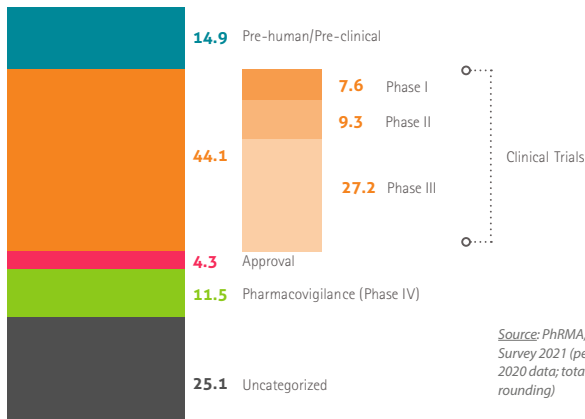


Source: SCRIP – EFPIA calculations (according to nationality of mother company)

Source: European Federation of Pharmaceutical Industries and Associations (2022)

Cost on R & D in Pharmaceutical industry

ALLOCATION OF R&D INVESTMENTS BY FUNCTION (%)



Source: PhRMA, Annual Membership Survey 2021 (percentages calculated from 2020 data; total values may be affected by rounding)

Source: European Federation of Pharmaceutical Industries and Associations (2022)

According to a recent review (Wong, Siah & Lo, Biostatistics, 2019), between 2000 and 2015

- **41.0%** of confirmatory clinical trials overall and
- **64.5%** of confirmatory clinical trials in oncology

have been unsuccessful.

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have been unsuccessful.

- **13.4%** of treatments entering Phase I receive approval
- In Oncology only **3.4%** of treatments entering Phase I receive approval

- Avoid going straight into large and expensive phase III trials
- Take more care during phases I and II
- Explore the potential of "new" statistical methods:
 - ▶ Sequential designs
 - ▶ Adaptive designs
 - ▶ Bayesian methods

Fixed Sample Designs



- no learning during conduct of trial
- clear separation between development phases

A general approach for fixed sample designs

- Parallel group comparison of efficacy
- E: Experimental, C: Control
- $n = n_E + n_C$
- Superiority trial (advantage for E sought)

H_0 : E no different from C vs H_1 : E superior to C

A general approach for fixed sample designs

θ : measure of advantage of E over C

$\theta > 0$: E superior

$\theta = 0$: No difference

$\theta < 0$: E inferior

θ_R : target measure of advantage of E over C
(target effect)

α : $\mathbb{P}\{\text{Reject } H_0 | \theta = 0\}$
Type I error rate (one-sided)

$1 - \beta$: $\mathbb{P}\{\text{Reject } H_0 | \theta = \theta_R\}$
Power to detect the target effect

Score Test

- $x_1, x_2, \dots$ each with density $f(x|\theta)$
- Log-likelihood of θ is

$$l(\theta|x_1, \dots, x_n) = \sum_{i=1}^n \log f(x_i|\theta)$$

- Expanding a Taylor's expansion gives

$$\begin{aligned} l(\theta) &= l(0) + \theta l'(0) + \frac{1}{2} \theta^2 l''(0) + \dots \\ &\approx \text{const} + \theta B - \frac{1}{2} \theta^2 V \quad (\text{for small } \theta) \end{aligned}$$

where $B = l'(0)$ and $V = -l''(0)$.

- As B is a sum $-\sum_{i=1}^n \frac{d}{d\theta} \log f(x_i|\theta)$ – it is asymptotically normally distributed.
- For large n and small θ

$$B \sim N(\theta V, V) \quad \text{and} \quad \frac{B}{\sqrt{V}} \sim N(0, 1)$$

- B is the efficient score statistic for θ
- For various response types B and V can be derived

Example: Normal data

treatment	E	C
mean	μ_E	μ_C
standard deviation (known)	σ	σ

- $\theta = \mu_E - \mu_C$
- $B = \frac{n_C n_E}{n\sigma^2} (\bar{X}_E - \bar{X}_C)$
- $V = \frac{n_C n_E}{n\sigma^2} = \frac{n}{4\sigma^2}$ for $n_E = n_C$

Example: Normal data

$$\begin{aligned} Z = \frac{B}{\sqrt{V}} &= \frac{\bar{X}_E - \bar{X}_C}{\sigma} \sqrt{\frac{n_C n_E}{n}} \\ &= \frac{\bar{X}_E - \bar{X}_C}{\sigma \sqrt{\frac{1}{n_E} + \frac{1}{n_C}}} \end{aligned}$$

And hence the Score test is equivalent to the Z-test in this situation.

$$\begin{aligned}\mathbb{P}\{B > c \mid \theta = 0\} &= \alpha \\ \mathbb{P}\left\{\frac{B}{\sqrt{V}} > \frac{c}{\sqrt{V}} \mid \theta = 0\right\} &= \alpha \\ 1 - \Phi\left(\frac{c}{\sqrt{V}}\right) &= \alpha \quad \text{as } \frac{B}{\sqrt{V}} \sim N(0, 1)\end{aligned}$$

so that $c = z_{1-\alpha}\sqrt{V}$, where z_γ is the γ percentile of the standard normal distribution.

Similarly

$$\begin{aligned}\mathbb{P}\{B > c \mid \theta = \theta_R\} &= 1 - \beta \\ 1 - \Phi\left(\frac{c - \theta_R V}{\sqrt{V}}\right) &= 1 - \beta\end{aligned}$$

as for $\theta = \theta_R$, $\frac{Z - \theta_R V}{\sqrt{V}} \sim N(0, 1)$.

Therefore

$$\begin{aligned}c - \theta_R V &= z_\beta \sqrt{V} \\ V &= \left(\frac{z_{1-\alpha} + z_{1-\beta}}{\theta_R}\right)^2\end{aligned}$$

when combining the results.

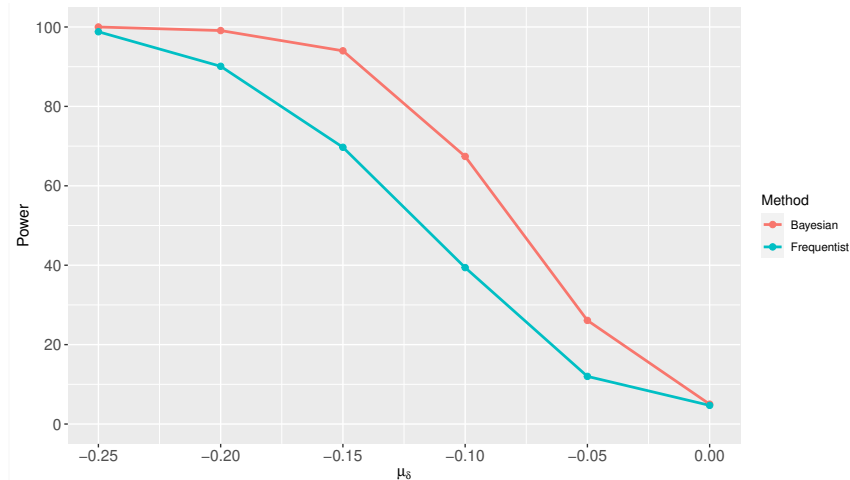
$$V = \left(\frac{z_{1-\alpha} + z_{1-\beta}}{\theta_R} \right)^2$$

- This formula has general validity
- V is a function of n and unknown parameters, which have to be estimated to obtain a sample size
- The relationship of V to n is the most approximate part of the procedure

When do traditional trials struggle: Rare diseases

- Neuroferritinopathy is an ultra-rare disease
- Fewer than 100 cases reported since its identification in 2001
- Some observational data available

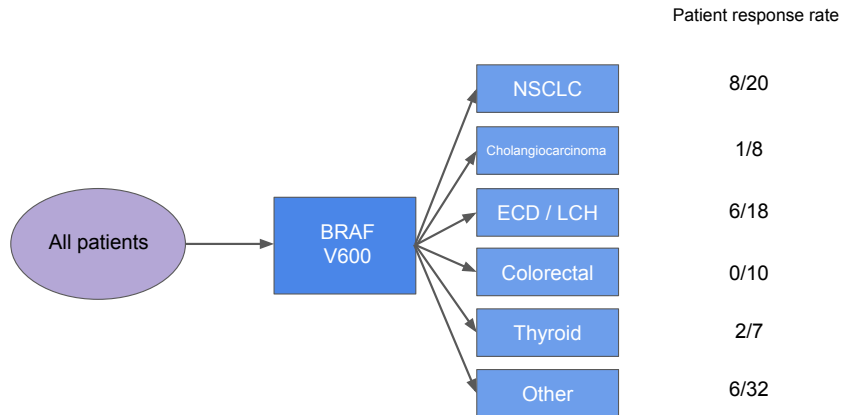
When do traditional trials struggle: Rare diseases



Basket trials in oncology – An example

Hyman *et al.* (2015) reported a recent basket trial, which has been designed to evaluate the efficacy of vemurafenib in BRAF-V600.

A total of 122 patients with BRAF-V600 mutations were enrolled, of which 95 entered the 6 modules.



References

-  Hyman D, Puzanov I, Subbiah V, et al. (2017) Vemurafenib in multiple nonmelanoma cancers with BRAF V600 mutations. New England Journal of Medicine. 373(8):726-736.
-  Wong CH, Siah KW, Lo AW (2019). Estimation of clinical trial success rates and related parameters. Biostatistics, 20:273-286.